



Hospital Management System (HealthSure)

Project Walkthrough

Presented by: Purnendu Shekhar Achary – Intern (1806521)

Module: Recipient

Date:

Project Overview: Streamlining Healthcare

The Hospital Management System (HMS) is a comprehensive digital solution designed to modernise and streamline hospital operations and healthcare workflows.



Core Purpose

To enhance efficiency, reduce administrative burden, and improve the overall patient experience.



Key Stakeholders

Designed to serve a diverse group including Patients, Doctors, Hospital Administrators, Insurers, and Pharmacists, ensuring a holistic approach to healthcare management.

System Modules: A Modular Approach

The HMS is built on a foundation of interconnected modules, each handling specific functionalities to ensure seamless operations.

Providers	Manages Doctor profiles, availability schedules, and specialisations.
Admin	Oversees system settings, user roles, and audit logs for security and compliance.
Recipient	Serves as the patient portal, facilitating insurance verification and provider search.
Insurance	Handles policy details, plan management, and claims processing.
Pharmacy	Manages prescription fulfilment and inventory.

Recipient Module: Patient-Centric Design

The Recipient module empowers patients with direct access to essential healthcare information and services.



Patient Portal

A user-friendly interface enabling patients to manage their healthcare journey efficiently.



Key Features

- View comprehensive insurance details.
- Search for healthcare providers based on various criteria.



Core Principles

Designed for simplicity, accuracy, and secure access, ensuring patient data integrity and ease of use.

My Contribution: Enhanced Recipient Features

My focus has been on developing and refining critical functionalities within the Recipient module, ensuring a robust and intuitive user experience.

1 View Insurance Details

Built an intuitive interface for viewing insurance details with search, filters, and a dedicated detail view page, ensuring smooth navigation, clear insights, and graceful error handling.

2 Search for Providers

Developed advanced search capabilities allowing filtering by name, specialisation, and location, ensuring patients find the right care provider.

3 Smart UI Logic & Validations

Integrated intelligent user interface logic and robust validation rules to guide users, prevent errors, and ensure data integrity.



Key Features: Recipient Module



View Insurance Details

Developed a robust and user-focused insurance view module where recipients can:

- Search by **plan name**
- Filter by **coverage status** (*All, Active, Expired*)
- Apply **date filters** to refine resultsIntegrated features like **sorting, pagination, and input validation** ensure a seamless user experience.
- Handled **system-level failures gracefully** using **custom exception handling**.
- Also built a **dedicated detail view page** to provide recipients with a clear, at-a-glance summary of their selected insurance policy.



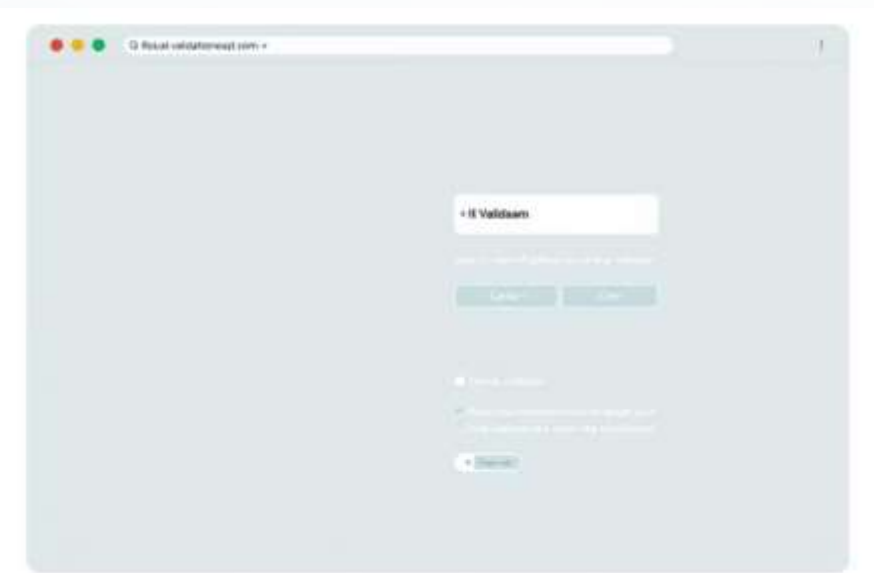
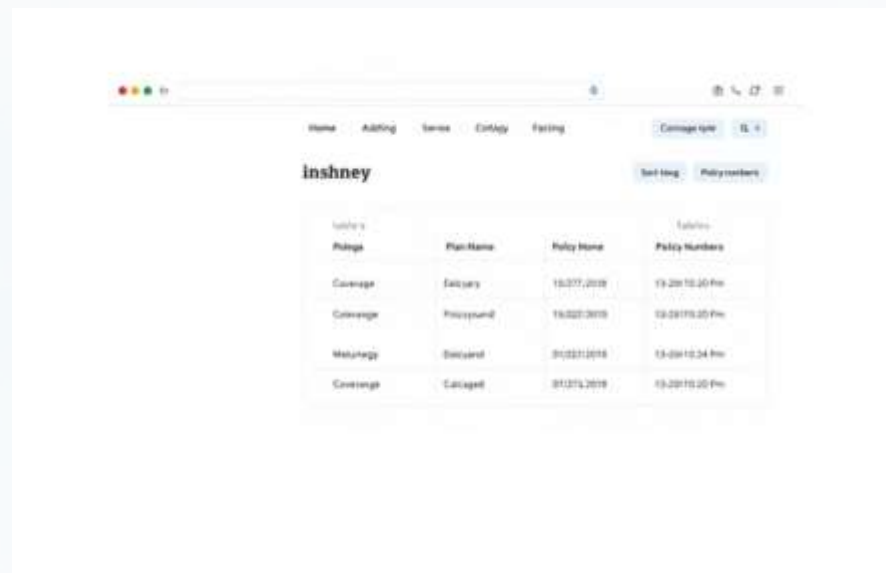
Search for Providers

Implemented an advanced, criteria-based doctor search allowing recipients to:

- Search by **doctor name** or **address**, with options for **Starts With** or **Contains**
- Select **specialization** from a **dynamically populated dropdown** (fetched from the database)
- Supports **sorting, pagination, and enterprise-grade validation** for input safety.
- Built-in **custom exception handling** ensures reliability during system issues.
- This module is also **integrated with appointment booking**, enabling users to find and connect with providers seamlessly.

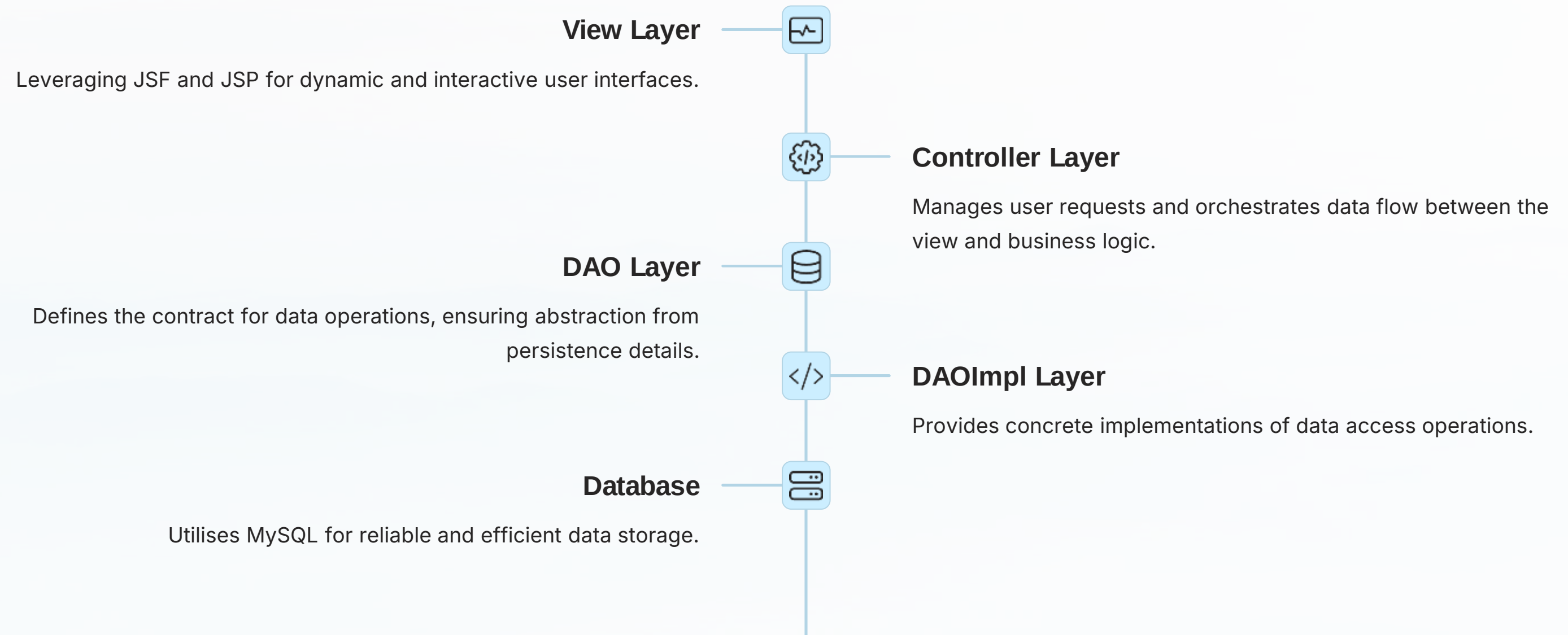
UI Screens: Visualising the Experience

These are illustrative examples of the user interface screens, showcasing the design and functionality implemented within the Recipient module. Actual screenshots will be presented during the live walkthrough.



Architecture and Tech Stack: Robust Foundations

The HMS is built on a layered architecture, ensuring maintainability, scalability, and a clear separation of concerns.



Key Principles: Custom exceptions for robust error handling, and strict separation of concerns for modular development.

The Core Technologies: A Visual Overview

The Hospital Management System leverages a robust and modern technology stack to ensure performance, reliability, and maintainability.



Java 1.8

The primary programming language powering the backend logic and application foundation.



Hibernate 3.6

An Object-Relational Mapping (ORM) framework for efficient and seamless database interaction (XML config).



Log4j

A logging framework for enabling robust debugging, auditing, and monitoring of application activities.



JSF 2.2 & JSP

Frameworks used for building dynamic, interactive, and robust user interfaces.



MySQL

The relational database management system chosen for reliable and scalable data storage.



System Reliability & Modularity

Achieved through custom exceptions for precise error handling and strict separation of concerns for maintainable and scalable development.

Wrap-Up & Code review

We've explored the modular design of the Hospital Management System and delved into the specifics of the Recipient module, highlighting its user-centric design and the clean backend layering.



Modular Design

Ensures scalability and ease of maintenance.



User-Centric

Prioritises ease of use and patient experience.



Clean Backend

Promotes robust and efficient data handling.

